Theory of Computation

Closure

Lecture 30

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Closure Properties

1) Union

2) Intersection

3) Concatenation

4) Difference

5) Reversal

6) Complement

7) Kleene Star

8) Subset. $L = [a^* - a^*]$ finite / Infinit

$(D, \circ) \Longrightarrow D.$

Closed.

$\circ \rightarrow$ operation

$D \rightarrow$ Type of Sets

D = Set of finite Strings

$L_1 \in D$

$L_2 \in D$

$L_1 \underset{\circ}{\cup} L_2 = L_3$

D = Infinite

a^*, b^*

a^+, b^+

$a^* b^1$

D = Set of Infinite lan

$\cup$	$\cap$	C	Difference $(-)$
$L_1 \cup L_2 = L_3$	$L_1 \cap L_2 =$ ___ not	$L_1^c =$ ε not	$L_1 - L_2 =$ ___ not
$\Downarrow$	$a^* \cap a^+ = a^+$	$\Sigma^* - L_1$	$a^* - a^+ = \varepsilon$
Closed Infinite	$a^* \cap b^* = \varepsilon$	$\Sigma^* - (aa)^*$ may or may not be Infi	$a^* - (aa)^* =$ ___
	may or may not be Infinite ϕ?		may or may not be Infi
Reversal	**Concatenation**	**Kleene Star**	**Subset**
$L_1^R =$ ___	$L_1 L_2 = L_3$	$L_1^* =$ Always Infinite	may or may not be
$a\Sigma^*$	Closed Always Infinite	Closed	not
$\Sigma^* a$ Infinite			

D = Rep lang.

$L_1 \cup L_2 =$

$L_1 = a^*$

$L_2 = (aa)^*$

$L_3 = (aaa)^*$

$(D, \circ) = D$

Reg $\cup$ CFL

$=$

Thank You